

## CHARLOTTETOWN, PE, Canada

WMO: 717060

Lat: 46.289N

Lon: 63.129W

Elev: 49

StdP: 100.74

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -19.3      | -17.1 | -25.4                         | 0.4 | -18.8 | -22.7 | 0.5 | -16.4 | 15.0                  | -0.2 | 13.4 | -1.0 | 4.6                   | 290  | 0.663 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.1                    | 27.0            | 20.8 | 25.6 | 19.8 | 24.2 | 19.0 | 22.0                | 25.4 | 21.0 | 24.1 | 20.1 | 23.0 | 5.3                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.8                            | 15.5 | 23.9 | 19.9 | 14.6 | 23.0 | 19.0 | 13.9 | 22.1 | 64.5          | 25.5 | 61.2 | 24.2 | 58.1 | 23.1 | 25.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 12.1              | 10.4 | 9.3 | -22.4                      | 29.8 | 2.5                | 1.3 | -24.2                                              | 30.7 | -25.7      | 31.5 | -27.1      | 32.2 | -28.8      | 33.2 |
|                   |      |     | -22.9                      | 23.6 | 2.4                | 0.8 | -24.6                                              | 24.2 | -26.0      | 24.7 | -27.4      | 25.2 | -29.1      | 25.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.3    | -6.7  | -6.5  | -2.5  | 3.3   | 9.3   | 14.7  | 19.3  | 18.9  | 14.9  | 9.0   | 3.4   | -2.3  |
|                                                                        | DBStd       | 10.00  | 5.83  | 5.38  | 4.59  | 3.89  | 3.90  | 3.84  | 2.74  | 2.81  | 3.60  | 3.80  | 4.48  | 4.93  |
|                                                                        | HDD10.0     | 2293   | 516   | 462   | 387   | 206   | 62    | 6     | 0     | 0     | 3     | 65    | 204   | 382   |
|                                                                        | HDD18.3     | 4511   | 774   | 696   | 645   | 452   | 281   | 120   | 22    | 28    | 114   | 289   | 449   | 640   |
|                                                                        | CDD10.0     | 943    | 0     | 0     | 1     | 4     | 39    | 146   | 289   | 274   | 150   | 34    | 5     | 0     |
|                                                                        | CDD18.3     | 119    | 0     | 0     | 0     | 0     | 1     | 11    | 53    | 44    | 10    | 1     | 0     | 0     |
|                                                                        | CDH23.3     | 484    | 0     | 0     | 0     | 0     | 6     | 52    | 218   | 176   | 33    | 1     | 0     | 0     |
|                                                                        | CDH26.7     | 53     | 0     | 0     | 0     | 0     | 0     | 3     | 23    | 23    | 4     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 4.6    | 5.0   | 4.9   | 5.0   | 4.9   | 4.5   | 4.2   | 4.0   | 3.8   | 4.1   | 4.5   | 4.9   | 5.1   |
| (15) Precipitation                                                     | PrecAvg     | 1217   | 112   | 94    | 98    | 92    | 94    | 94    | 83    | 88    | 96    | 112   | 119   | 132   |
|                                                                        | PrecMax     | 1579   | 263   | 176   | 191   | 178   | 218   | 202   | 232   | 210   | 178   | 215   | 216   | 259   |
|                                                                        | PrecMin     | 848    | 32    | 29    | 45    | 14    | 26    | 20    | 9     | 9     | 23    | 29    | 40    | 64    |
|                                                                        | PrecStd     | 173    | 40    | 35    | 34    | 35    | 44    | 42    | 47    | 49    | 39    | 47    | 39    | 47    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 10.4  | 8.4   | 11.3  | 17.8  | 24.2  | 26.8  | 28.9  | 29.0  | 26.5  | 21.3  | 16.8  | 11.9  |
|                                                                        |             | MCWB   | 9.8   | 7.6   | 7.6   | 12.7  | 16.1  | 20.0  | 22.0  | 21.5  | 21.0  | 16.9  | 15.2  | 11.3  |
|                                                                        | 2%          | DB     | 6.5   | 5.4   | 7.8   | 14.3  | 21.2  | 24.8  | 27.1  | 26.9  | 23.9  | 18.6  | 14.7  | 9.2   |
|                                                                        |             | MCWB   | 6.0   | 4.5   | 6.1   | 10.6  | 14.9  | 18.6  | 21.1  | 20.7  | 19.4  | 15.7  | 13.5  | 8.6   |
|                                                                        | 5%          | DB     | 3.9   | 3.0   | 5.7   | 11.8  | 18.4  | 23.1  | 25.8  | 25.2  | 22.1  | 16.7  | 12.7  | 6.7   |
|                                                                        |             | MCWB   | 3.0   | 2.0   | 4.0   | 9.0   | 13.1  | 17.6  | 20.1  | 19.7  | 18.3  | 14.5  | 11.5  | 5.9   |
|                                                                        | 10%         | DB     | 1.4   | 1.2   | 3.7   | 9.6   | 16.1  | 21.2  | 24.4  | 23.8  | 20.5  | 15.0  | 10.3  | 4.3   |
|                                                                        |             | MCWB   | 0.6   | 0.2   | 2.2   | 6.8   | 12.0  | 16.6  | 19.3  | 19.0  | 17.5  | 12.8  | 8.9   | 3.3   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 9.8   | 7.9   | 8.9   | 14.5  | 17.8  | 21.8  | 23.3  | 23.1  | 21.7  | 18.2  | 15.6  | 11.5  |
|                                                                        |             | MCDB   | 10.4  | 8.1   | 10.2  | 16.4  | 21.5  | 25.7  | 27.3  | 26.7  | 24.7  | 19.9  | 16.5  | 11.9  |
|                                                                        | 2%          | WB     | 6.2   | 4.6   | 6.5   | 11.8  | 15.9  | 20.1  | 22.2  | 21.8  | 20.4  | 16.4  | 13.8  | 8.8   |
|                                                                        |             | MCDB   | 6.6   | 5.2   | 7.5   | 13.5  | 19.3  | 23.3  | 25.9  | 25.0  | 22.8  | 18.0  | 14.6  | 9.2   |
|                                                                        | 5%          | WB     | 3.0   | 2.1   | 4.3   | 9.5   | 14.2  | 18.7  | 21.1  | 20.8  | 19.3  | 14.9  | 11.7  | 6.0   |
|                                                                        |             | MCDB   | 3.6   | 2.8   | 5.5   | 11.2  | 17.2  | 21.7  | 24.3  | 23.9  | 21.4  | 16.5  | 12.5  | 6.6   |
|                                                                        | 10%         | WB     | 0.7   | 0.2   | 2.3   | 7.1   | 12.5  | 17.3  | 20.2  | 19.9  | 18.0  | 13.1  | 9.1   | 3.5   |
|                                                                        |             | MCDB   | 1.2   | 1.0   | 3.5   | 9.2   | 15.5  | 20.3  | 23.3  | 22.7  | 20.0  | 14.8  | 10.1  | 4.2   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 7.8   | 7.7   | 7.0   | 7.6   | 9.2   | 8.9   | 8.1   | 8.1   | 8.0   | 6.9   | 6.3   | 6.2   |
|                                                                        |             | MCDBR  | 11.4  | 9.3   | 9.3   | 11.4  | 13.2  | 11.4  | 10.0  | 9.7   | 9.3   | 8.7   | 8.9   | 9.2   |
|                                                                        |             | MCWBR  | 11.8  | 9.2   | 8.2   | 8.3   | 8.0   | 6.8   | 5.4   | 5.1   | 5.9   | 6.7   | 8.6   | 9.7   |
|                                                                        | 5% WB       | MCDBR  | 11.4  | 9.5   | 9.0   | 10.1  | 11.3  | 9.9   | 8.4   | 8.3   | 8.3   | 8.2   | 8.8   | 9.2   |
|                                                                        |             | MCWBR  | 11.9  | 9.5   | 8.6   | 8.8   | 7.8   | 6.7   | 5.3   | 5.0   | 6.0   | 7.2   | 9.0   | 9.8   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.299 | 0.289 | 0.318 | 0.348 | 0.365 | 0.381 | 0.407 | 0.393 | 0.355 | 0.333 | 0.312 | 0.289 |
|                                                                        | taud        |        | 2.343 | 2.306 | 2.321 | 2.396 | 2.415 | 2.388 | 2.353 | 2.387 | 2.491 | 2.528 | 2.505 | 2.465 |
|                                                                        | Ebn at Noon |        | 792   | 887   | 908   | 910   | 901   | 884   | 855   | 854   | 862   | 833   | 783   | 775   |
|                                                                        | Edh at Noon |        | 75    | 94    | 108   | 111   | 113   | 116   | 119   | 111   | 92    | 76    | 65    | 60    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.08  | 1.79  | 3.15  | 4.19  | 5.08  | 5.53  | 5.63  | 4.97  | 3.81  | 2.35  | 1.22  | 0.83  |
|                                                                        | RadStd      |        | 0.07  | 0.24  | 0.29  | 0.38  | 0.47  | 0.42  | 0.38  | 0.32  | 0.33  | 0.24  | 0.15  | 0.07  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | +0.52 | +0.71       | N/A     | N/A     | N/A     |
| (47) Regional (42 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | +0.35       | N/A     | N/A     | N/A     |

Nomenclature: See separate page